

YICHUN HEILONGJIANG, China

WMO: 507740

Lat: 47.7081N

Lon: 128.8358E

Elev: 259

StdP: 98.25

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.0	-30.8	-35.9	0.1	-32.9	-33.7	0.2	-30.5	7.6	-15.6	6.6	-15.8	0.4	0	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.1	20.8	28.7	20.3	27.2	19.8	23.1	27.3	22.2	26.4	21.3	25.2	2.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.8	17.0	25.3	20.8	15.9	24.4	19.9	15.1	23.5	70.2	27.4	66.7	26.5	63.4	25.3	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.4	5.4	-35.8	33.1	2.4	1.7	-37.5	34.4	-38.9	35.4	-40.2	36.4	-41.9	37.6		
			-35.9	24.5	2.4	1.0	-37.6	25.2	-39.0	25.7	-40.3	26.3	-42.0	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-20.8	-15.5	-6.1	4.6	12.1	18.1	21.4	19.4	12.8	3.9	-8.7	-18.9
	DBStd	15.45	4.48	5.24	6.15	4.58	4.04	3.36	2.31	2.76	3.81	4.72	6.70	4.93
	HDD10.0	4031	955	714	499	171	22	0	0	0	16	198	559	896
	HDD18.3	6177	1213	947	758	411	197	44	3	20	170	449	809	1154
	CDD10.0	1091	0	0	0	10	88	242	353	291	99	7	0	0
	CDD18.3	196	0	0	0	1	4	37	98	53	3	0	0	0
	CDH23.3	1699	0	0	0	6	121	433	730	367	41	2	0	0
	CDH26.7	427	0	0	0	1	22	132	200	67	4	0	0	0
Wind	WSAvg	2.0	1.6	1.8	2.3	2.6	2.4	1.9	1.8	1.6	1.8	2.1	1.9	1.7
Precipitation	PrecAvg	643	4	5	9	21	53	100	165	163	74	29	12	7
	PrecMax	995	11	16	23	63	132	206	346	341	130	76	33	22
	PrecMin	455	0	0	2	2	14	21	60	22	5	2	1	1
	PrecStd	129	3	4	5	13	30	39	71	73	31	20	8	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.5	3.0	12.3	23.6	29.1	32.9	32.4	30.2	27.1	21.6	11.2	0.4
		MCWB	-7.1	-1.2	5.2	11.2	15.9	20.5	22.3	22.2	19.0	13.2	6.1	-2.0
	2%	DB	-8.9	-0.2	8.9	19.8	26.6	29.8	30.5	28.6	24.2	17.4	7.2	-6.1
		MCWB	-10.4	-3.3	2.7	9.4	14.8	19.3	21.6	21.5	17.0	10.5	3.2	-7.4
	5%	DB	-11.0	-3.1	6.1	16.9	24.2	28.0	28.9	27.0	22.2	14.6	3.9	-8.6
		MCWB	-12.1	-5.4	1.2	8.1	13.9	18.6	21.5	20.7	15.9	8.8	1.0	-9.7
	10%	DB	-12.9	-6.0	3.5	14.0	21.5	25.8	27.3	25.5	20.2	12.1	1.1	-10.8
		MCWB	-13.9	-7.9	-0.4	6.4	12.3	18.1	21.0	20.0	14.7	7.2	-1.1	-11.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.9	-1.0	5.7	12.3	17.3	22.3	24.4	24.2	20.4	14.0	6.6	-1.9
		MCDB	-5.6	2.6	11.7	21.3	25.5	29.3	28.5	28.2	25.2	19.7	10.3	0.2
	2%	WB	-10.3	-3.2	3.3	10.2	16.0	20.9	23.4	22.8	18.5	11.4	3.4	-7.3
		MCDB	-9.1	-0.4	8.0	18.7	24.0	27.2	27.7	26.6	22.0	16.0	6.7	-6.2
	5%	WB	-12.1	-5.5	1.5	8.5	14.6	19.9	22.6	21.8	17.0	9.5	1.1	-9.7
		MCDB	-11.0	-3.1	5.5	15.8	22.2	25.6	26.9	25.4	20.5	13.9	3.6	-8.7
	10%	WB	-13.9	-8.0	-0.2	6.9	13.3	18.8	21.8	20.8	15.7	7.7	-1.1	-11.7
		MCDB	-12.9	-6.1	3.2	13.3	20.0	23.8	26.0	24.1	19.1	11.5	1.1	-10.8
Mean Daily Temperature Range	5% DB	MDBR	13.0	14.8	12.9	12.5	13.3	12.0	10.1	10.2	12.7	11.6	11.0	11.3
		MCDBR	13.8	15.9	14.2	17.5	18.3	16.4	13.5	12.9	15.9	16.4	13.3	11.9
		MCWBR	12.4	13.2	9.3	9.4	8.4	6.9	5.9	6.3	8.4	9.6	9.8	10.7
	5% WB	MCDBR	13.7	15.7	13.3	15.8	15.5	12.9	10.8	10.5	12.2	14.7	12.7	11.6
		MCWBR	12.4	13.0	8.8	9.0	7.6	6.0	5.5	5.6	7.0	9.3	9.6	10.5
Clear-Sky Solar Irradiance	taub	0.295	0.322	0.392	0.493	0.545	0.534	0.514	0.457	0.421	0.412	0.343	0.301	
	taud	2.163	2.083	1.959	1.842	1.792	1.916	2.024	2.179	2.182	2.061	2.157	2.142	
	Ebn at Noon	758	815	805	757	734	747	757	786	775	697	683	690	
	Edh at Noon	84	114	153	191	209	186	165	135	122	117	85	76	
All-Sky Solar Radiation	RadAvg	1.92	3.03	4.33	4.76	5.17	5.40	5.06	4.68	3.91	2.68	1.81	1.54	
	RadStd	0.09	0.12	0.33	0.51	0.50	0.73	0.54	0.35	0.38	0.20	0.14	0.08	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+52	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page